Jinyu Zheng

New York, NY | jinyu.zheng.5.5@gmail.com | 1-929-992-8922 | zhengjinyu | jzheng40

Education

New York University — Expected *May 2028*

B.S. in Electrical Engineering

- GPA: 3.25/4.00
- **Coursework:** Data Structures and Algorithms, Analog Electronics, PCB Design, Embedded Systems, Signals and Systems, Machine Learning

Projects

STM32F405 Dev Board - *Altium Designer, PCB Design, Power Electronics*

- Designed an STM32F405 ARM Cortex-M4 development board in Altium Designer integrating SPI/I^2C peripherals and SWD debugging.
- Implemented a 24V to 3.3V dual-stage power delivery network (buck converter + LDO) featuring power multiplexing, TVS ESD, and reverse polarity protection.
- Configured PCB layer stackup and routed an impedance-matched USB-C interface using differential pair length matching, continuous ground planes, and via stitching.
- Executed DRC/DFM checks, generated Gerber/BOM files for JLCPCB fabrication, and validated power rails during board bring-up using a DMM and oscilloscope.

Water Level Controller - *Bare-metal C++, Embedded System, Controls*

- Developed bare-metal C++ firmware for an ATmega32U4 MCU to operate a real-time, closed-loop physical control system.
- Programmed microcontroller registers directly to establish precise 1kHz Fast PWM control signals and configure continuous analog-to-digital conversions.
- Mitigated actuator chattering and hardware overshoot by engineering a hybrid Proportional/Hysteresis control loop and mathematically modeling physical transition delays.

Ragdoll Simulator - *Python, Physics, Modeling* — Glowscript Link

- Engineered a Python/VPython physics engine simulating gravity, rotational kinematics, and momentum in real time for an interactive rag-doll simulator.
- Implemented custom collision detection and environmental physics, including restitution (bounce), dynamic friction, and air resistance to ensure mathematically accurate object interactions.

Experiences

Audio Visual Technician @ KAPCQ CM — *Sept 2020 - Present*

- Configured microphone arrays, PA systems, and wireless transmitters for live events and presentations.
- Operated multi-channel audio mixers, balancing signal levels and preventing clipping, distortion, and feedback during live operation.
- Diagnosed signal issues like RF interference, grounding noise, and signal loss in audio systems during events.
- Ensured system readiness by verifying cabling, signal routing, and equipment functionality prior to events.

Skills

Languages: Python, Java, C, C++, C#, MATLAB
Software: Altium, Git, Microsoft Office, Google Workspace
Hardware: Oscilloscope, DMM, Bench Power Supply
Embedded: GPIO, UART, SPI, I^2C, ADC, PWM